Time-Resolved Elemental Analysis of Catalysts in a Fixed-Bed Reactor by Laser-Induced Breakdown Spectroscopy and Chemometrics

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Introduction

Currently, *in-situ* characterization techniques provide molecular, surface, electronic, and structural information about a catalyst inside a reactor. However, one piece of information that is hardly accessible yet is the change in the "*Elemental Composition*" of a catalyst material under reaction conditions.[1] Considering that catalyst deactivation by coking or poisoning and the modulation of catalysts by promoters or inhibitors is a common challenge in industrial catalysis. This represents a significant gap in the field of *in-situ* characterization. In this context, a novel laser spectroscopy technique, *Laser-Induced Breakdown Spectroscopy (LIBS*), might contribute to closing this gap due to its high sensitivity, multi-element detection capability, and versatility for remote operation.

Materials and Methods

To assess the suitability of LIBS for detecting changes in elemental composition remotely, a fiber-optic LIBS system (FO-LIBS) was developed and coupled with a fixed-bed reactor. We conducted three cycles of deactivation and regeneration, measuring elemental, activity, and temperature profiles. The study focused on propane dehydrogenation (PDH) over a VO_x/Al_2O_3 catalyst, known for its strong deactivation due to coke formation[2]. Chemometric analysis further enhanced our results, allowing for quantitative insights into catalyst composition.

Results and Discussion

Overall, the carbon LIBS profiles correlated well with the gas phase measurements obtained during periodic operation. As catalysts deactivate, their dehydrogenation activity decreases exponentially. *In-situ* LIBS reveals the elemental catalyst composition over time, with coke deposits detected after 30 minutes, and continuously increasing until a plateau is reached. During regeneration, carbon deposits are burned into CO_x, causing the LIBS carbon signal to decrease. Chemometric analysis effectively uses LIBS spectra data by correlating the amount of carbon burned as CO_x

with the LIBS spectra obtained at various times during the regeneration phase. A Partial Least Squares (PLS) model was developed, which demonstrated good regression statistics. However, the calibration strategy limits the model's application to the regeneration phase, leading to underestimation of carbon deposits during deactivation, although trends remain acceptable

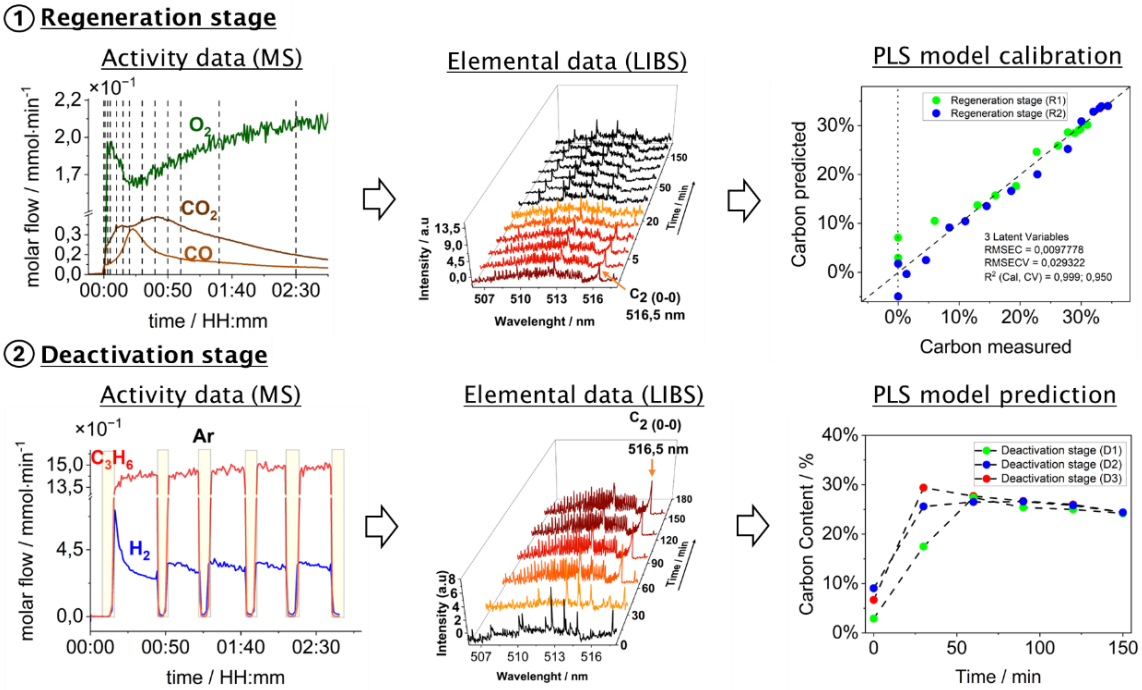

Figure 1. Chemometric analysis of species and elemental temporal profiles obtained during *in-situ* LIBS.

Significance

A cost-effective fiber-optic LIBS setup was developed for examining hard-to-reach areas such as fixed-bed reactors. It enables front and side sampling with spatial and time resolutions of 700 µm and seconds, respectively, which is enough for most catalytic applications. The invasiveness and the sensitivity to the gas atmosphere have been recognized as the most critical parameters limiting universal application. Despite these challenges, there is potential for the *in-situ* LIBS methodology to be adapted for different types of reactor configurations, including fluidized-bed and multiphase reactors. The characteristic flow patterns in these reactors will continuously refresh the sampling position, and probing of the liquid phase will provide the additional advantage of homogeneity at a molecular level.

References

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